

Education

Emory University BA in Computer Science, Economics & Mathematics (GPA: 3.54)

Dec 2017

Skills

Languages

Java, Python, JavaScript, TypeScript, Go (Golang), SQL, C#

Tools

Docker, Kubernetes, PostgreSQL, MySQL, SQL Server, DynamoDB, Neo4j, AWS, S3, Databricks, PySpark (Spark), Airflow, Kinesis, Thrift, Terraform, FastAPI, Pydantic, Alembic, React, GraphQL, Cypher, OpenTelemetry, Prometheus, Grafana, Bazel

Systems

Distributed systems, Entity Resolution, Knowledge Graphs, Data Pipelines (ETL), Agentic LLM Workflows (GenAI data quality), REST, microservices, IAM, access control, observability, DevOps

Work Experience

Dataminr

Senior Software Engineer

Sep 2025 - Present

- Led platform engineering on Knowledge Platform 2.0 across a 4-engineer team. Greenfield PostgreSQL entity resolution and knowledge graph system replacing a Neo4j-based platform serving 10M+ entities and 19M+ relationships, supporting 39 entity types and 10+ data sources. Powers real-time alerting for 1,000+ enterprise and government customers.
- Built agentic data-quality pipelines: LLM workflows ingesting human content-review signals to auto-populate missing entities and enrich existing records in production
- Built Bronze/Silver data layer architecture: entity identity model partitioned by entity type, temporal versioning for point-in-time entity views, and authority-based conflict resolution across 4 trust tiers (auto-resolve & human-review thresholds).
- Designed and shipped the bulk-write submission API: S3-backed payload storage, async submission endpoints, per-row idempotent writes, and resumable job state. Enabled migration of millions of entities from 5+ legacy data sources
- Unifying Dataminr's fragmented knowledge graphs into a single consolidated graph — the core architectural mandate of KP 2.0: built ETL pipelines (Databricks/PySpark) and entity resolution integration for cyber intelligence sources; designing schema and ontology reconciliation to ingest billions of entities from the ThreatConnect acquisition
- Redesigned the graph's data model to make Person a first-class entity — previously buried as nested attributes on Organizations — to power executive-protection alerting: owned the architecture, led a junior engineer through implementation

Meta

Software Engineer

Apr 2022 - Oct 2024

- Architected and implemented scalable role-based access control for Meta's internal governance platform (PIMS) and replaced coarse, manual gating with dynamic, granular permissions; reduced access request burden by 71% across eng, legal, and policy teams; directed technical execution with support from 2 engineers
- Led cross-functional design and implementation of Assessor Reporting export infrastructure in PIMS, and enabled batch PDF/CSV exports across apps, violations, enforcements, and disputes; cut audit prep time by 52% in partnership with Legal

Oracle

Oracle Cloud Infrastructure (OCI) *Software Engineer*

Jan 2020 – Feb 2022

- Built and maintained a tier-zero distributed Limits system (Java, Terraform, Go) enforcing IAM-scoped quotas across 150+ services in 40+ global regions; designed policy-aware APIs and region build automation to provision 250+ cloud regions including U.S. government environments, reducing manual setup by 90%.

Alarm.com

Software Engineer

Jun 2018 – Nov 2019